

CeramTec RK 87 Alumina, 99.8%**Categories:** [Ceramic](#); [Oxide](#); [Aluminum Oxide](#)**Material Notes:** Aluminas exhibit good mechanical properties such as hardness, compressive and tensile strength, and elastic modulus. They perform well at elevated temperatures.**Key Words:** Al₂O₃, aluminum oxide**Vendors:** No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	3.96 g/cc	0.143 lb/in ³	DIN EN 623-2
Water Absorption	0.00 %	0.00 %	Open Porosity; DIN EN 623-2
Permeability	0.00	0.00	%, Gas
Weibull Modulus	15	15	DINV ENV 843-5

Mechanical Properties	Metric	English	Comments
Vickers Microhardness	2000	2000	HV1; DINV ENV 843-4
Tensile Modulus	406 GPa	58900 ksi	Young's; DINV ENV 843-2
Flexural Strength	630 MPa	91400 psi	DIN EN 843-1
Compressive Strength	4000 MPa	580000 psi	DIN 51067T1
Poissons Ratio	0.23	0.23	DINV ENV 843-2
Fracture Toughness	4.30 MPa-m ^{1/2}	3.91 ksi-in ^{1/2}	K _{IC} (SEVNB); DIN CEN/TS 14425-1
Shear Modulus	165 GPa	23900 ksi	Calculated

Electrical Properties	Metric	English	Comments
Volume Resistivity 	5.00e+8 ohm-cm @Temperature 400 °C	5.00e+8 ohm-cm @Temperature 752 °F	IEC 672-1
	5.00e+14 ohm-cm @Temperature 20.0 °C	5.00e+14 ohm-cm @Temperature 68.0 °F	IEC 672-1
Dielectric Constant	9.0 @Frequency 1e+6 Hz	9.0 @Frequency 1e+6 Hz	IEC 672-1
Dielectric Strength	19.0 kV/mm	483 kV/in	IEC 672-1
Dielectric Loss Index	0.0050 @Frequency 9.00e+9 Hz	0.0050 @Frequency 9.00e+9 Hz	IEC 672-1

Thermal Properties	Metric	English	Comments
CTE, linear 	7.50 µm/m-°C @Temperature 20.0 - 400 °C	4.17 µin/in-°F @Temperature 68.0 - 752 °F	DIN EN 821-1
	8.50 µm/m-°C @Temperature 20.0 - 1000 °C	4.72 µin/in-°F @Temperature 68.0 - 1830 °F	DIN EN 821-1
Specific Heat Capacity	0.900 J/g-°C	0.215 BTU/lb-°F	DINV ENV 821-3
Thermal Conductivity	30.0 W/m-K	208 BTU-in/hr-ft ² -°F	DIN EN 821-2
Maximum Service Temperature, Air	1500 °C	2730 °F	
Maximum Service Temperature, Inert	1500 °C	2730 °F	

Component Elements Properties	Metric	English	Comments
Al2O3	99.8 %	99.8 %	
Descriptive Properties			
Ra = Arithmetic Mean Roughness Value (µm)		<0.05	
Thermal Shock Resistance R1 (K)		159	calculated; $R1 = [\sigma_s (1-\mu)] / (\alpha \cdot E)$

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.